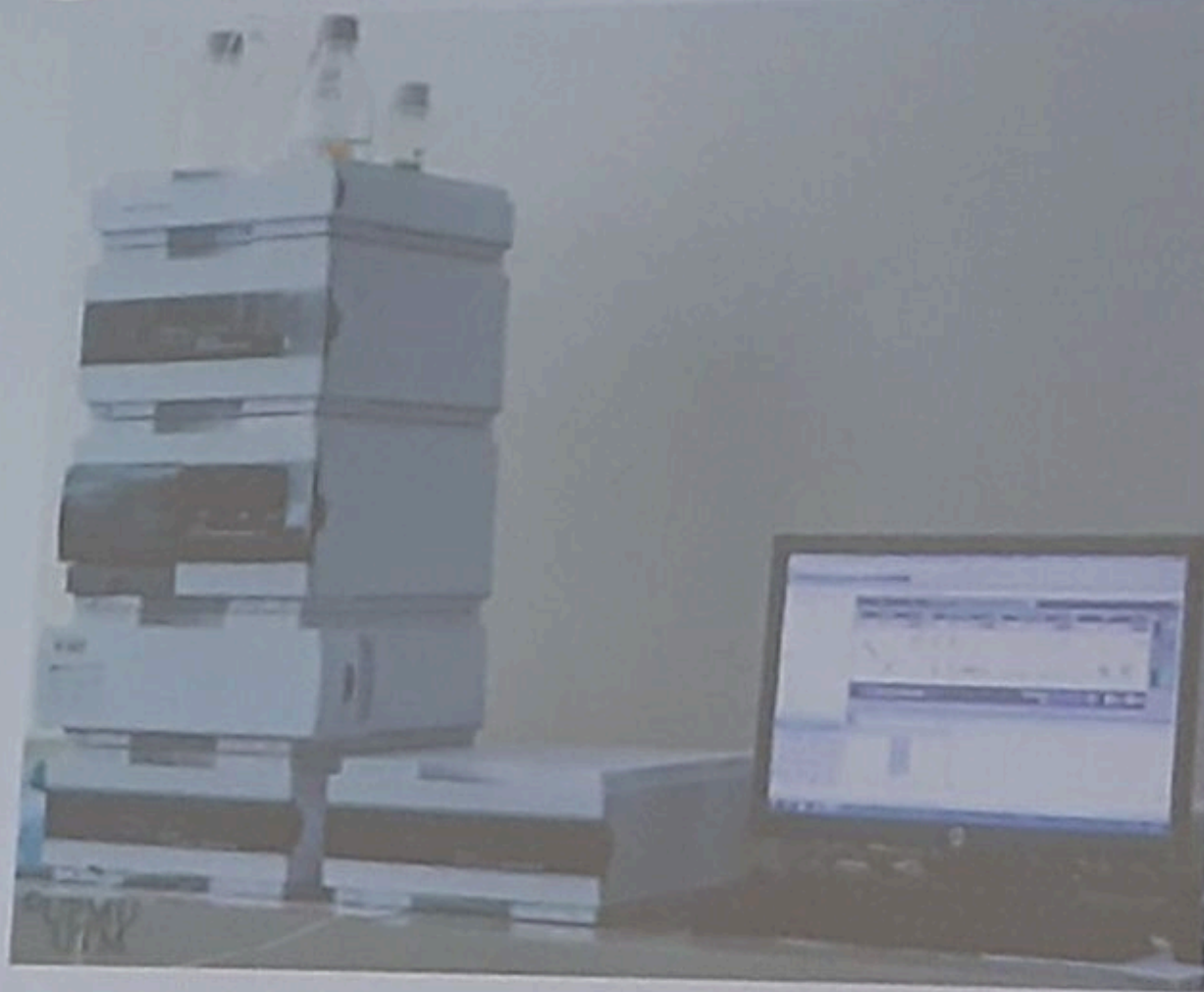


Modern chromatography

High Performance Liquid Chromatography (HPLC)

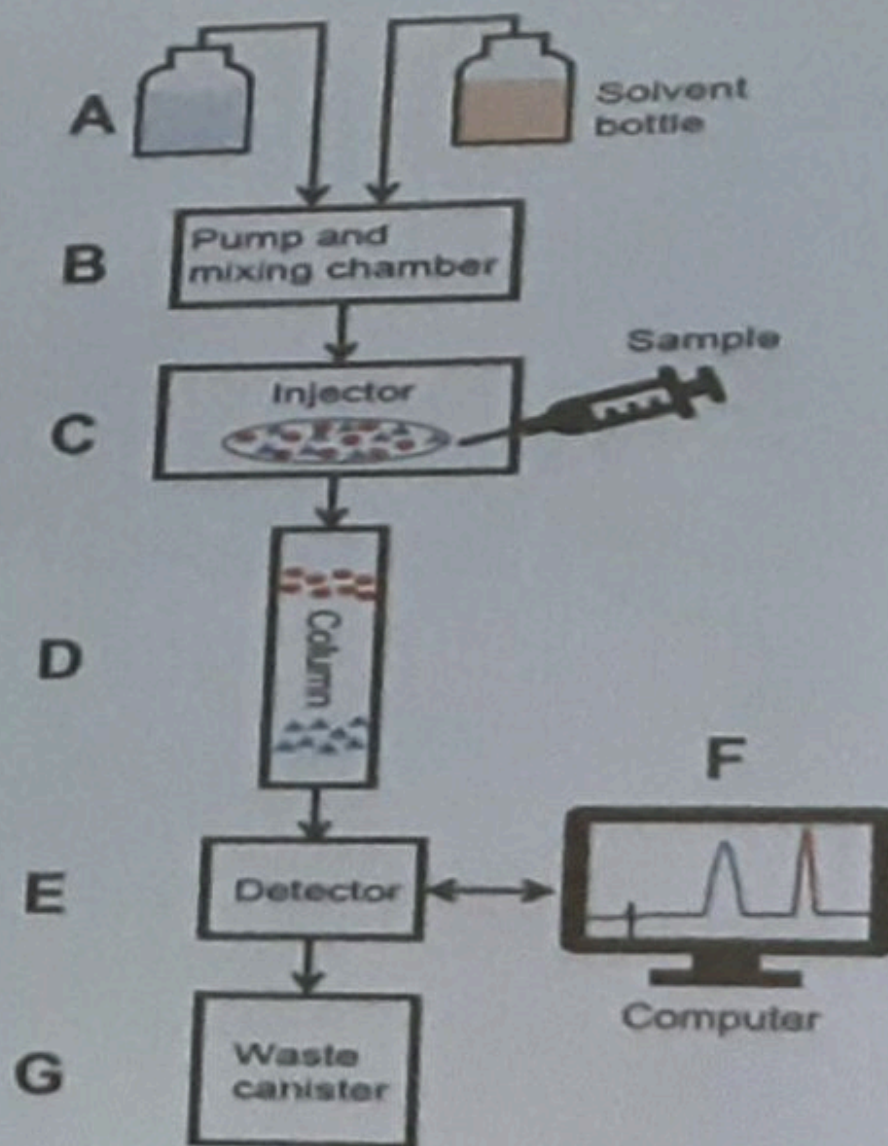
- HPLC is a column chromatography technique where the **mobile phase flows under extremely high pressure**, provided by special pumps.
- It utilizes a very finely packed stationary phase with **particle size $< 150\mu\text{m}$**



HPLC system setup

Components of HPLC:

- Solvent reservoir (**A**)
- Pump (**B**)
- Injection device (**C**)
- Guard column and separation column (**D**)
- Detector (**E**)
- Computer (**F**)
- Sample collector (**G**).



Mobile phase (Solvent system)

- Like most other chromatographic systems should be inert, low viscosity, efficient, appropriate and cost effective.
- The solvent depends on SP:
 - **Normal phase:** non-polar solvents such as hexane or iso-octane
 - **Reverse phase:** polar solvents such as water, acetonitrile or methanol.
- The HPLC system can be set up either for isocratic or gradient elution.
- Use only HPLC grade solvent to avoid noisy baselines
- Dissolved air should be removed (Degassing) using passing the solvent through a vacuum
- In a UV detector the UV absorbance of the solvent should not interfere with sample's UV range

The pump

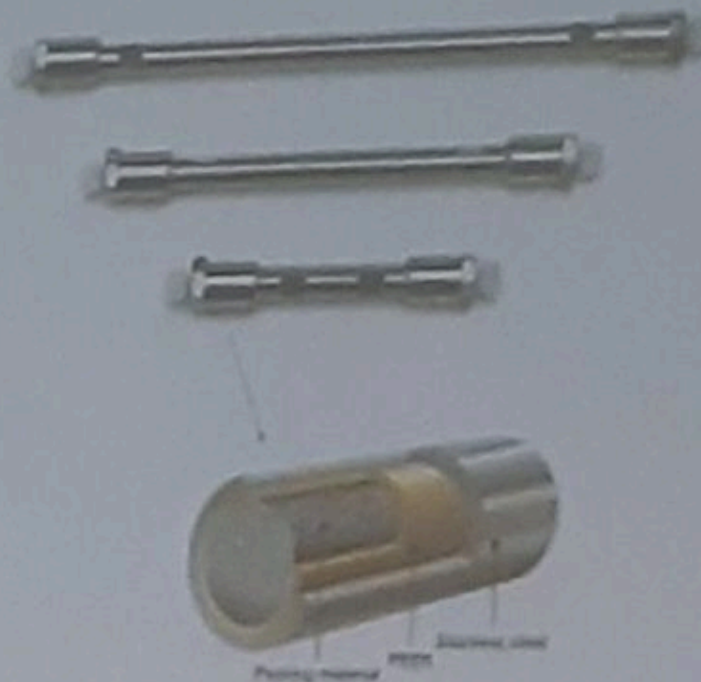
- Purpose is to provide a **constant flow of mobile phase** through the chromatographic column.
- Considerations concerning pump:
 - The component of the pump must be **resistant to** corrosive chemicals and solvents, such as acids, bases and aggressive organic solvents.
 - Flow should be **pulse free and stable**.

Injection system

- Can either be manual or automated (autosampler)
- Introducing the sample to the flowing mobile phase at the head of the column
- Withdraws programmed amount of sample from sample container (vial) and injects it in the loop system

The Column

- Separation in HPLC occurs in the column
- The column is the most important component and determines the performance and resolution of HPLC system.
- It is usually constructed of stainless steel and packed with an adsorbent or a liquid phase bonded to a stationary support.
- Most HPLC columns are typically 25 cm long with an internal diameter of 4.6 mm
- The particle size varies particle diameter is in the range of 3-10 μm .
 - The most popular particle size for analytical column is 5 μm .



Common stationary phase used in HPLC

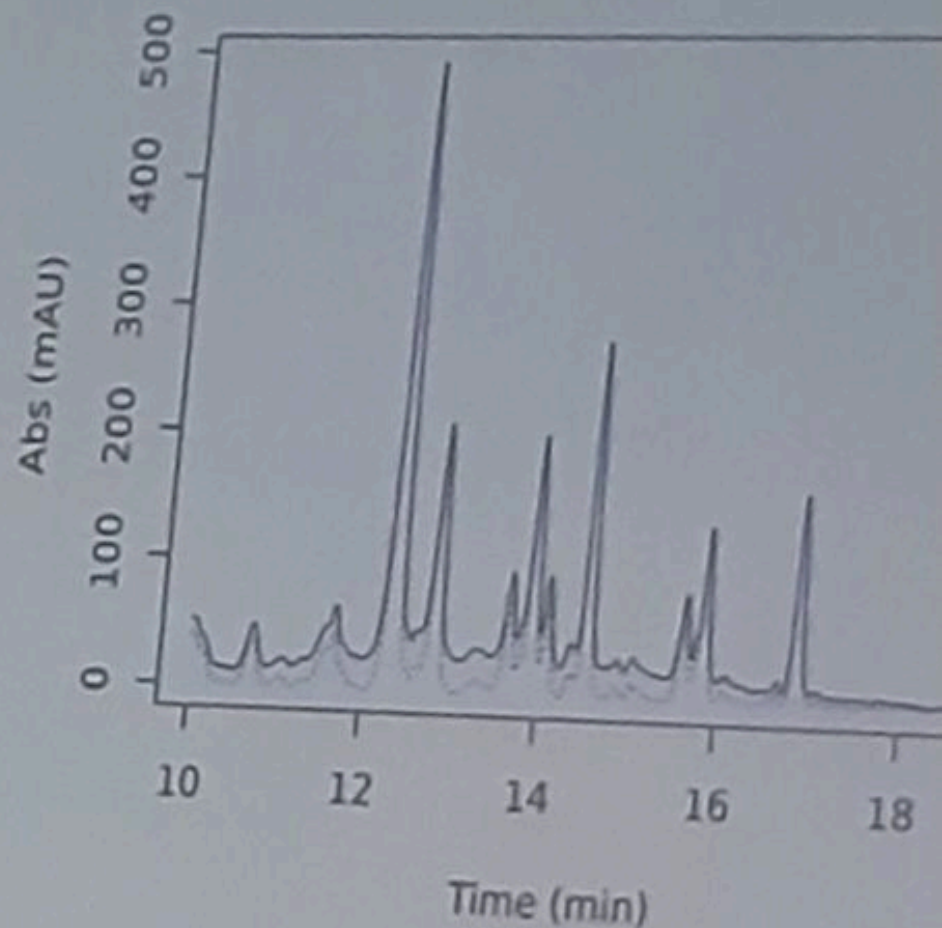
- **Adsorption HPLC:** silica or alumina
- **Partition HPLC:** solid-bound phases, usually on silica or alumina
 - **Normal phase:** amino ($-\text{NH}_2$), cyano ($-\text{CN}$), diol groups
 - **Reverse phase:** C-2 ($\text{Si}-\text{CH}_2\text{CH}_3$), C-8 ($\text{Si}-(\text{CH}_2)_7\text{CH}_3$) or C-18 ($\text{Si}-(\text{CH}_2)_{17}\text{CH}_3$).
- **Size exclusion HPLC:** gels of different type.
- **Ion exchange HPLC:** ion-exchange resins.

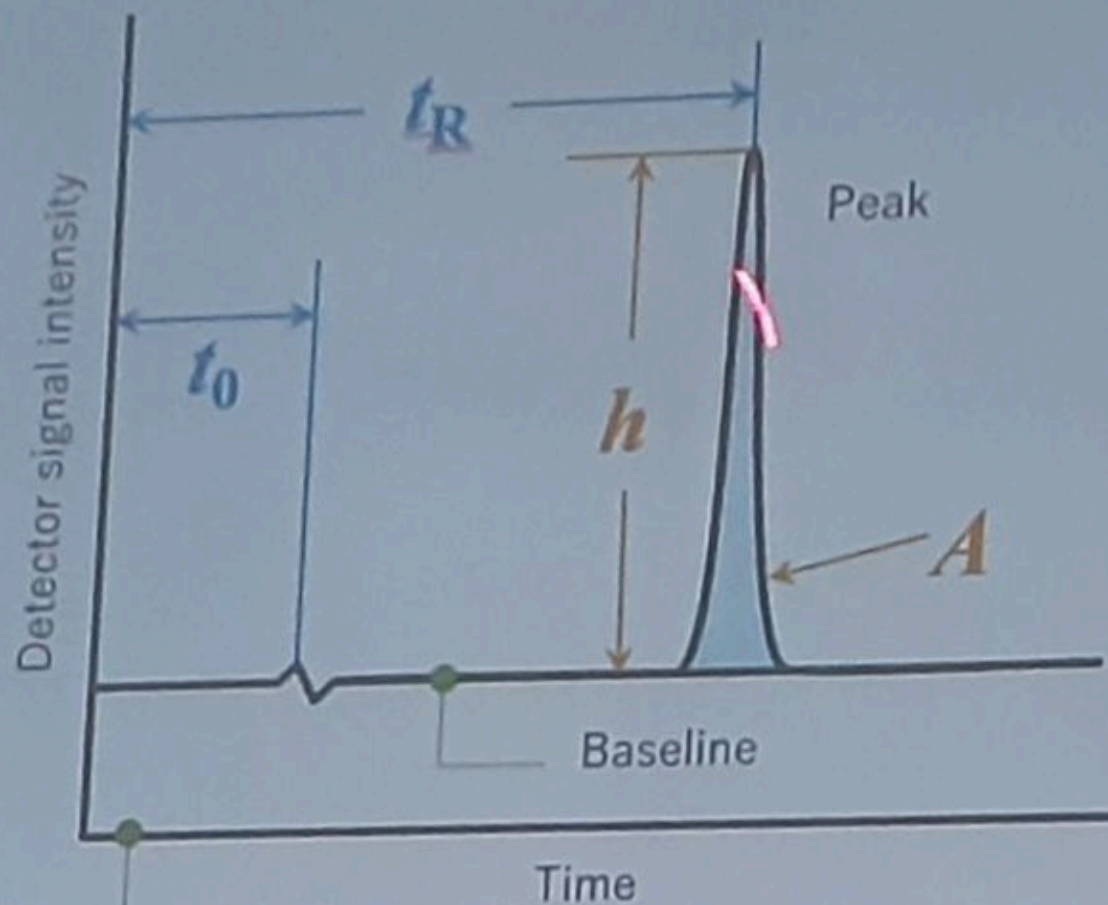
HPLC detectors

- Selection of a detector depends on the specific nature of the solute
- There are four commonly used HPLC detectors.
 - Fluorescence detector
 - Electrochemical detectors
 - Refractive index detector
 - UV/visible detector (specific UV scanner and DAD)

HPLC chromatogram

- It's a graph of signal intensity plotted against retention time
- A solute that has passed through the detector are translated in to a single peak
- Portrays a qualitative and quantitative (area under the curve) data
- The tallest peak is usually the major compound



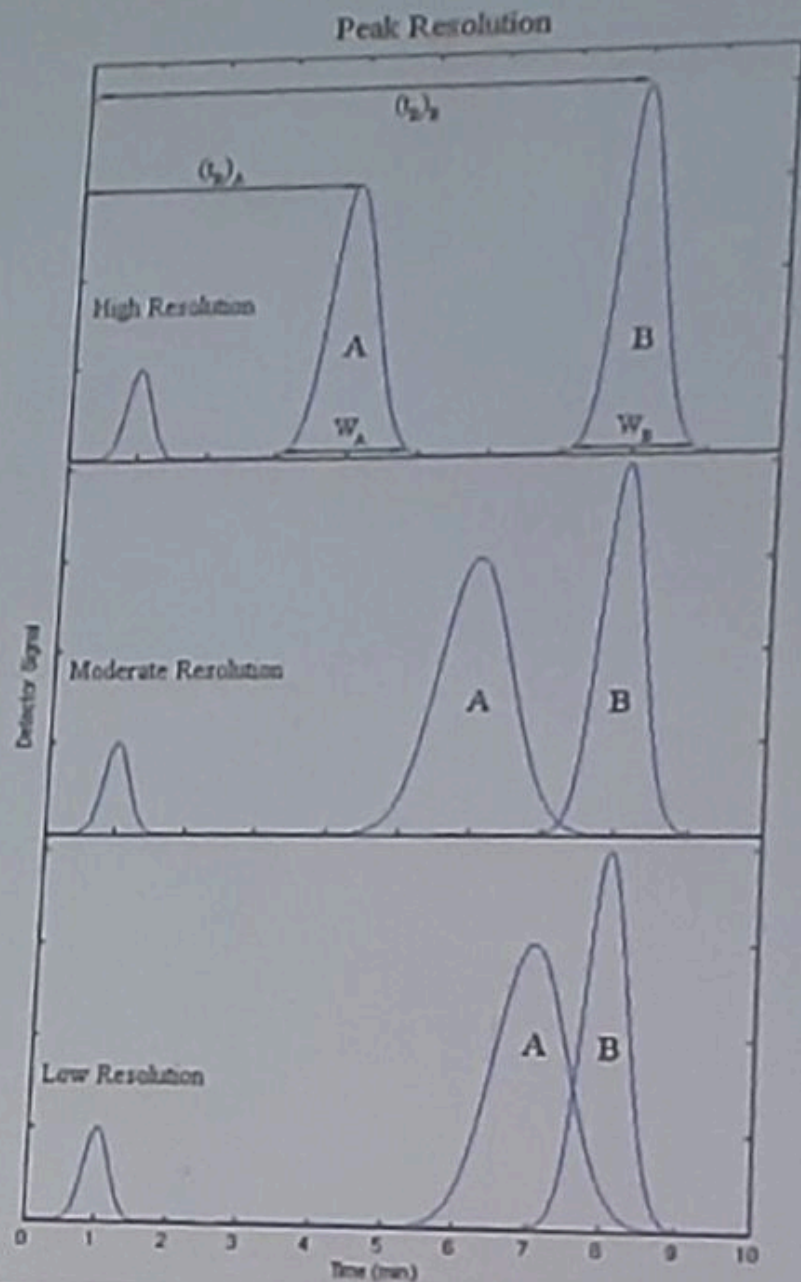


- t_R : Retention Time
- t_0 : Dead Time
- A : Peak Area
- h : Peak Hight

Injection of the sample into the column → Start of analysis

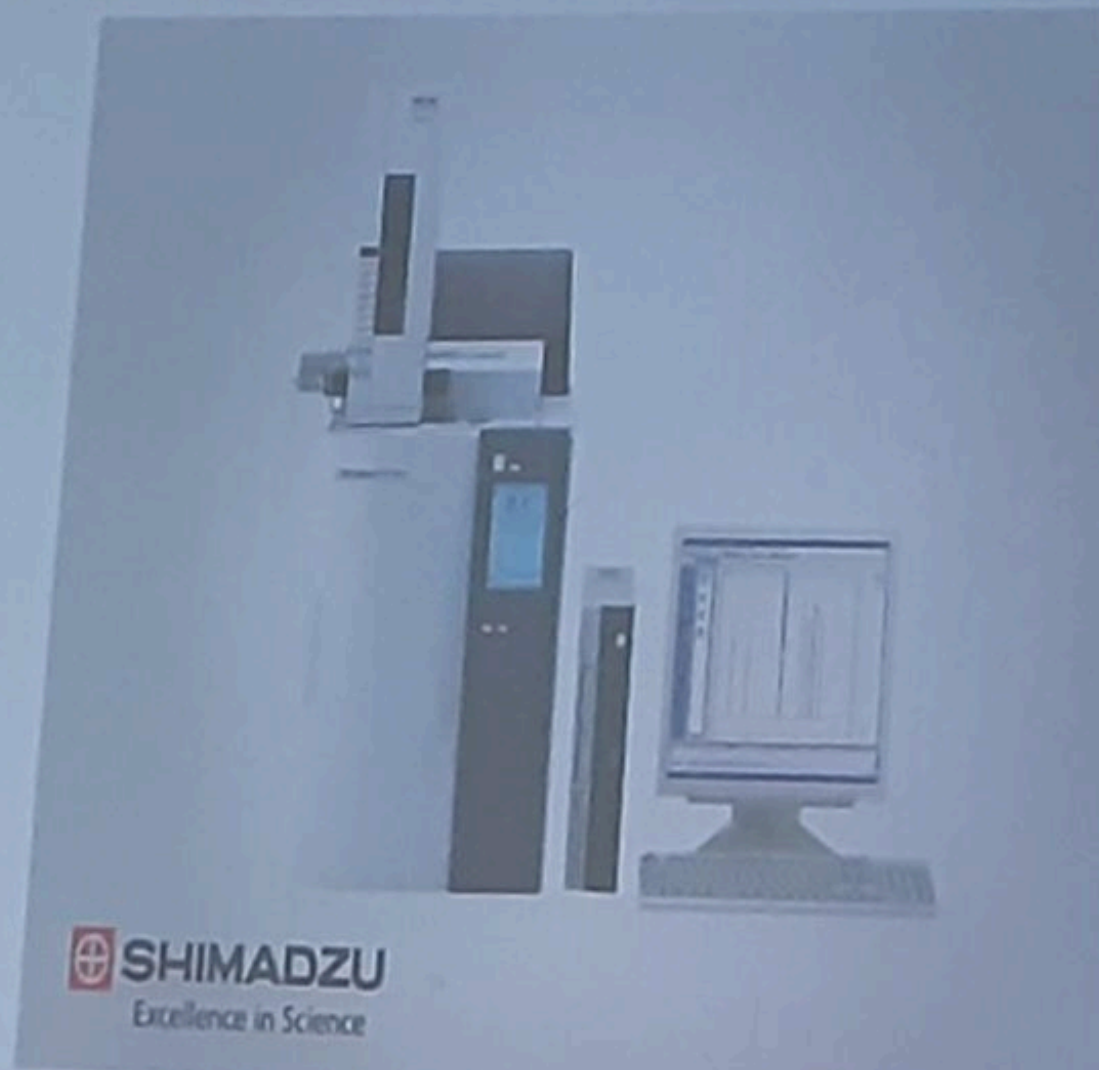
Factors affecting HPLC performance

- **Injection volume (sample concentration):** high concentration would lead to low resolution (tailing)
- **Column dimensions:** thinner column and long column gives better resolution
- **Particle size of the SP:** smaller particle size -> better resolution
- **Flow rate of solvent:** low flow rate give better resolution
- **Polarity:** Solvent, SP and Solute
- **Temperature:** higher temperature higher resolution



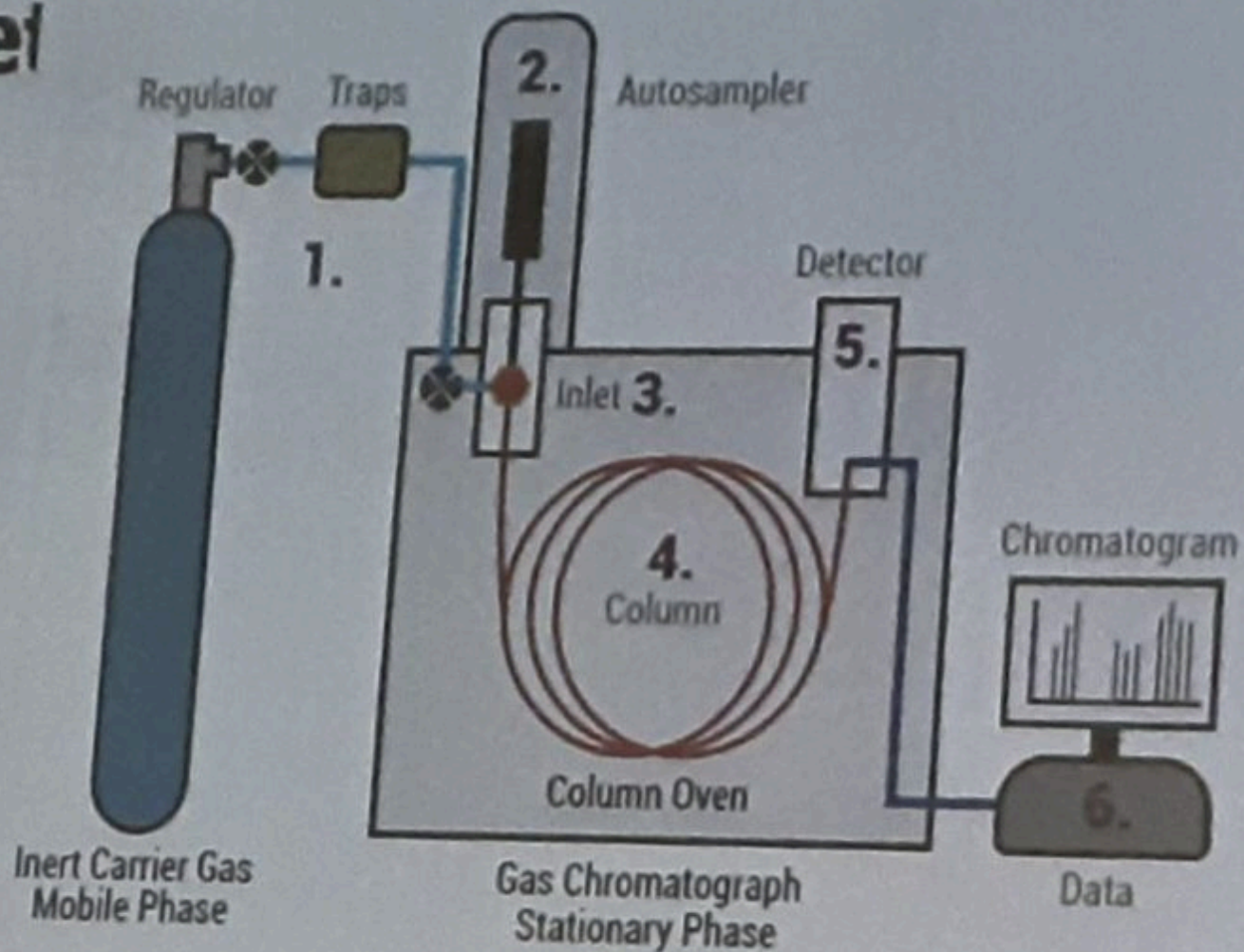
Gas chromatography (GC)

- Gas chromatography is that form of chromatography in which a gas is the mobile phase.
- It is used to analyze volatile substances
 - Eg: volatile oil(essential oils), derivatized compounds
- If the stationary phase is a:
 - Solid GC technique is called gas – solid chromatography (GSC)
 - if it is a liquid, the technique is called gas – liquid chromatography (GLC).



GC system set

1. Carrier gas
2. Auto sampler
3. Injection port
4. Column in an oven
5. Detector
6. Computer to generate a chromatogram



How it works ?

- The sample is vaporized and carried by the mobile gas phase (the carrier gas) through the column.
- Separation is effected by competition between for the stationary phase and volatility at the column temperature
 - That is retention forces Vs partial vapour pressure in the mobile phase.
- Samples partition (equilibrate) into the stationary liquid phase, based on their solubilities at the given temperature.
- The solutes or analytes separate from one another based on their relative vapor pressures and affinities for the stationary bed.
- In GSC separation depends on the adsorption profile of the solute.

What types of compounds are suitable for GC analysis?

- Compounds that possess appreciable **volatility** at temperatures below 350–400 °C.
- As a general rule, the **greater the molecular weight or polarity** of a compound, **the lower its volatility**.
- compound must be able to **withstand high temperatures** and be rapidly transformed into a vapor **without degradation or reacting** with other compounds.
- Some compounds need to be derivatized:
 - Because they are not **sufficiently volatile**
 - To increasing **thermal stability**
 - To **increase detector response** by incorporating into the derivative functional groups which produce a higher detector signal
 - To improve **separation and reduce tailing**.

Mobile gas in GC

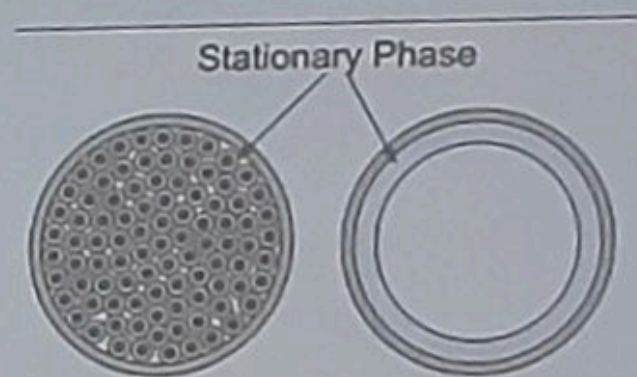
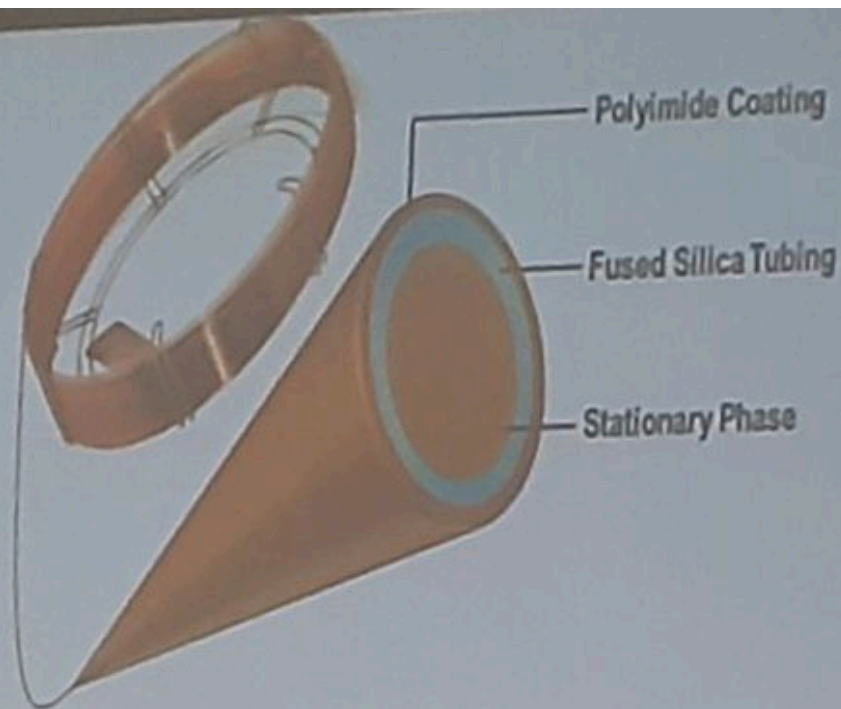
- High purity gases are supplied from a pressurized cylinder or gas generator.
- Pressure regulators on the cylinders or generators control the amount of gas delivered to the gas chromatograph.
- Flow controllers maintain the **flow-rate accurately** over all the full operating temperature range of 0—450°C.
- Since at constant pressure the flow-rate will change, as **the viscosity of most gases varies with temperature**.

Sample injector (Autosampler)

- Gas sampling methods require that the entire sample be in the gas phase under the conditions in use.
- The injection port should be hot enough to vaporize all the sample rapidly.
- But not too hot so that thermal decomposition or chemical rearrangement is avoided.
- Samples can be injected automatically with mechanical devices that are often placed on top of gas chromatographs.

GC column

- GC columns are usually made of copper or glass for the outer layer.
- Two types packed and capillary
- packed column generally has an internal diameter of 2-10 mm and a length of 1-4 m.
- Capillary column internal diameter is 0.1-0.5 mm and may have a length of 10-100 m.



	Typical packed column	Typical capillary column
Internal diameter	2-10 mm	0.1-0.5 mm
Stationary phase layer thickness	1-10 μm	0.1-8 μm
Length	1-4 m	10-100 m
Capacity	10 $\mu\text{g/peak}$	100 ng/peak
Peaks obtained	tend to be short and wide	tend to be long and sharp
		
Advantages	• Lower cost • Easy to make • Larger samples	• Higher separation efficiency • Faster separation • Better for complex mixtures

Oven

- The **column temperature is controlled** so that a good separation will occur in shorter time.
- The column temperature should be **high enough** so that sample components pass through it at a reasonable speed.
- At higher temperatures, the retention times decrease.
- Lower temperature means longer analysis times, but better resolution.
- **Isothermal vs. Programmed temperature**
 - **Isothermal** denotes a chromatographic analysis at one constant column temperature.
 - **Programmed temperature** refers to a linear increase of column temperature with time

Detectors

- Both qualitative and quantitative analysis
- The most common detector is the **flame ionization detector (FID)**.
 - It has the desirable characteristics of **high sensitivity, linearity, and detectivity** and yet it is relatively simple and inexpensive.
 - The column effluent is **burned in a small oxy-hydrogen flame** producing some **ions in the process**.
 - These ions are collected and form a **small current that becomes the signal**.
 - When no sample is being burned, there should be little ionization, the small current arising from impurities in the hydrogen and air supplies.

Advantages and disadvantages

Advantages

- Fast analysis, typically minutes
- Efficient, providing high resolution
- Sensitive, easily detecting ppm and often ppb
- Nondestructive, making possible on-line coupling; e.g., to mass spectrometer
- Highly accurate quantitative analysis
- Requires small samples, typically μL
- Reliable and relatively simple
- Inexpensive

Disadvantages

- Limited to volatile samples
- Not suitable for thermally labile samples
- Fairly difficult for large, preparative samples